CREATIVE TECHNO COLLEGE, ANGUL
PROJECT SYNOPSIS

1. Project Topic: Simple Drawing App Using Python
2. Group members:
    - Asutsoh Sahu
    - Biswajit Sahu
    - Debasis Nayak
    - Himanshu Shekhar Sahoo
    - Kunmun Sahu
    - Omm Prakash Mohanty
3. Course: BCA 1st Sem
4. Year: 2025-2029
5. Guide Name: Mr. Saumya Ranjan Pradhan & Mr. Swastik Ranjan Sahoo
6. Submission Date: 30-01-2026
7. Project Overview: (Explain in 150 – 200 words)

- The **Simple Drawing App** is a basic Python application developed using the **Tkinter** library. It allows users to draw freely on a digital canvas using the mouse. The interface is simple and user-friendly, making it easy for beginners to understand and use.
- The app uses mouse events like clicking, dragging, and releasing to draw on the canvas. It includes tools such as a pencil for freehand drawing and an eraser to remove mistakes. Users can also change brush color and thickness, and use undo and redo features to correct errors easily.
- Additionally, the application allows users to clear the canvas and save their drawings as image files using the **Pillow (PIL)** library. This project helps students learn GUI development, event handling, and basic drawing concepts in Python.

8. Objective: Clearly state the goals of the project

- To create a simple drawing application using Python and Tkinter.
- To allow users to draw on a canvas using the mouse.
- To provide basic tools like pencil and eraser.
- To allow users to select colors and change line thickness.
- To add undo, redo, and clear canvas options.
- To save the drawing as an image file.
- To help students learn GUI programming and mouse event handling.

9. Methodology: Provide a step-by-step plan for how the group intends to complete the project

- Create the main window and canvas using Tkinter.
- Add buttons for pencil, eraser, colour selection, undo, redo, clear, and save.
- Use mouse events to start drawing, draw while moving the mouse, and stop drawing.
- Store drawing actions to support undo and redo functionality.
- Use a colour chooser to change drawing colours and a slider to adjust line thickness.
- Save the canvas as an image using the Pillow (PIL) library.
- Test the application to ensure all tools work properly.

10. References:

- Python Software Foundation, Tkinter Documentation, Pillow (PIL) Documentation, YouTube, Chrome

Student Signatures

Sign. Of Guide with date